σ-Additive Probability on Orthomodular Lattices: A Survey and an Open Problem

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Abstract

The passage from a finitely coherent assignment of probabilities to a countably additive measure is, in the Boolean case, governed by a single condition (countable additivity) and executed by classical machinery (Carathéodory; Loomis–Sikorski; Stone duality). On a *non-distributive* orthomodular lattice (OML) — the algebra of propositions of a quantum system — this machinery breaks, and the question of when finite coherence forces countable behaviour is open. We survey the problem at the level needed to begin work on it. After fixing the apparatus (orthomodular lattices, the gap between orthogonality and meet-zero, the state/meet-additive/charge ladder, the McDonald–Bimbó dual space), we separate the question into two axes. The *extension* axis is classical (Horn–Tarski feasibility) and fully settled: on $L(H)$ no state of any kind extends, by an elementary finite construction; the *descent* axis is genuinely open and rests on a representation that does not exist: a Loomis–Sikorski theorem for σ-complete OMLs. The stake is more than technical: such a representation would furnish a *point-free*, σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory — built from the entailment relation rather than descended from a sample space. We review what is known and what is ruled out, and close with the two directions in which the open problem might be approached.

1 Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \varnothing \quad \Longrightarrow \quad \mu(A_n) \to 0.$$

De Finetti [7] held that only finite additivity is empirically grounded; Kolmogorov [23] adopted σ-additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements that can be performed together — is Boolean. Gluing contexts along their shared observations (a categorical colimit) returns a Boolean algebra when the contexts are mutually compatible, but a generically non-distributive *orthomodular lattice* (OML) when they are not: Gunji et al. [16] show the gluing produces an OML, and the compatible/incompatible dichotomy is the organising point of the companion paper [24]. The OML is forced by the context structure, not postulated; incompatibility — quantum complementarity, but equally contextual experimental design or interfering sensors — is what drives the observation algebra out of the Boolean world. For the non-quantum examples the incompatibility must be *operationally imposed* (the contexts genuinely non-co-realizable, with no joint probability space): it cannot arise intrinsically from the dynamics of a single classical system, since any finite family of classical observables on a common space jointly distributes (Kolmogorov), hence stays Boolean — contextuality there is failure of a global section (Abramsky–Brandenburger [1]; Fine [11]), a statement about non-co-realizable settings rather than about one underlying process. The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On any such OML distributivity may fail, so Carathéodory's outer measure — which needs a Boolean algebra of sets — does not apply. For the single lattice $L(H)$ this is rescued by Gleason's rigidity theorem [14], but Hilbert space is special: for general OMLs no analogue is known. Dropping distributivity does not by itself remove the 2-valued homomorphisms the Boolean engine needs — MO_3 is non-distributive yet still carries them (§2.1, §4) — but it removes the *guarantee*, and for $L(H)$ at dim ≥ 3 they fail outright (Kochen–Specker [22]). Whether enough survive is a question of concreteness, not non-distributivity, and it is the qualifier on which the open problem turns (§4, §5).

This note asks the shared question — *when does finite coherence force countable behaviour?* — in the OML setting. Our organising observation is that the question, vague as usually posed, splits into two axes that the Boolean case fuses: an *extension* axis, classical and fully settled (on $L(H)$ no state extends), and a *descent* axis that is genuinely open. The survey is built around that split.

The relational standpoint. Two established moves motivate the approach. De Finetti's operationalism [7] founds probability without a presupposed sample space: probabilities are coherent commitments about events, not measures on a pre-given Ω. Quantum theory, since Birkhoff and von Neumann, reads propositions as a lattice — the orthomodular lattice of §2 — rather than as subsets of a phase space. Combining the two, we begin from the propositions and their entailment order, not from a space Ω of outcomes, and ask what that relational structure alone determines. This is the measure-theoretic, non-distributive counterpart of pointless (localic) topology, which recovers spaces from their lattices of opens. The natural dual is then not a space of points but the McDonald–Bimbó space of *filters* (§2.3), and a positive answer to the open problem would be a *point-free* σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory (§5). The standpoint is developed further in [25].

2 Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1 Orthomodular lattices, and the orthogonal/meet-zero gap

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^{\perp}$ satisfying, for all a, b:

(i) $a \wedge a^{\perp} = \mathbf{0}$ and $a \vee a^{\perp} = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^{\perp} \leq a^{\perp}$ (order-reversing).

Definition 2.2 (Orthogonal; meet-zero)**.** Elements a, b are *orthogonal*, written $a \perp b$, if $a \leq b^{\perp}$ (equivalently $b \leq a^{\perp}$) — one entails the other's negation, *decisively incompatible*. They are *meet-zero* if $a \wedge b = \mathbf{0}$ — no common refinement, a weaker condition.

> **The gap.** Orthogonality always implies meet-zero; the converse holds in every Boolean algebra but **fails** once the lattice is non-distributive. *Meet-zero is strictly weaker than orthogonal* — the distinctions developed below all turn on this.

Definition 2.3 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \le b \quad \Longrightarrow \quad b = a \vee (a^{\perp} \wedge b). \tag{1}$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^{\perp}$ the orthogonal complement.

Example 2.4 (MO_n, and the gap made concrete)**.** MO_n is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. *Picture n lines through the origin: each line a is $\perp$ to its perpendicular $a^{\perp}$, but two atoms from distinct pairs are non-perpendicular lines.* It is orthomodular and non-distributive ($n \ge 2$): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad \text{(meet only at the origin)} \qquad \text{yet} \qquad a \not\perp b,$$

so meet-zero $\neq$ orthogonal already here. MO_3 is the smallest non-distributive OML.

Definition 2.5 (OMP; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.3: joins $a \vee b$ are required only for *orthogonal* pairs. An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins — i.e. its propositions can be modelled as actual *sets of outcomes*. (Concreteness is the property that separates the open frontier from the settled cases in §5, so it earns a name now.)

Remark 2.6 (Concreteness does not close the gap)*.* A set-representable OMP needs only closure under complement and *disjoint* union [3]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so orthogonality ($A \cap B = \varnothing$) implies meet-zero, but not conversely. Indeed MO_3 is *itself* concrete (it carries an order-determining set of two-valued states; Gudder [15]), so the paradigmatic meet-zero-$\neq$-orthogonal example is a concrete logic. The condition that closes the gap is not concreteness but a richness/atomicity condition (meet-zero coincides with orthogonality iff every nonempty $A \cap B$ contains a nonzero L-member); the abstract form $a \wedge b = \mathbf{0} \Rightarrow a \perp b$ is Tkadlec's *Boolean orthoposet* condition [34] (which does *not* mean Boolean *algebra*).

2.2 The state/meet-additive/charge ladder

Definition 2.7 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0,1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$. *(Probability, but additivity is only guaranteed where propositions are decisively incompatible.)* Three strengthenings, differing in *which* pairs additivity is demanded on, strictly increasing in general (e.g. on MO_3, Example 2.8):

(1) *State* — additive on orthogonal pairs.
(2) *Meet-additive* — additive on every *meet-zero* pair $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a) + v(b) = v(a \wedge b) + v(a \vee b)$, a different condition.
(3) *Charge-extendible* — the n-ary condition $\sum_i s(a_i) \le 1$ for every pairwise-meet-zero family $\{a_i\}$ (strictly stronger again). This is the *extension* condition of §3.

Example 2.8 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^{\perp}), (q, q^{\perp}), (r, r^{\perp})$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

2.3 The McDonald–Bimbó dual space

Definition 2.9 (Dual space; representation map)**.** The McDonald–Bimbó duality [26] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) \subseteq F(A)$ is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^{\dagger}$. Define $\mu(h(a)) = s(a)$. *(Read $h(a)$ as "the points where a holds" and μ as the state pushed onto those sets — the open problem is whether μ survives countable operations on them.)*

Remark 2.10 (The duality is finitary — the wall)*.* The identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for *finite* joins but fails for countable ones: finitary OML homomorphisms need not preserve infinite suprema. This finitariness is the wall the open problem of §5 runs into. (It is a *countable*-join phenomenon, not specific to joins: since $a \mapsto a^{\perp}$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^{\perp})^{\perp}$, so the failure to preserve countable meets is its De Morgan dual — the wall obstructs countable meets and joins together.) The rival OML duality of Cannon–Döring [4] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4 The Boolean engine: Stone duality

In the Boolean case, the descent argument ("σ-additive $\Leftrightarrow$ the measure concentrates on the physical points") runs on *Stone duality*: a Boolean event algebra embeds in the clopen algebra of its Stone space, a charge extends to a Baire measure there (Carathéodory), and σ-additivity is equivalent to concentration of that measure on the realised points [25]. That σ-additive step is realised measure-theoretically by the meagre-sets characterisation of σ-additivity (Rao–Rao [32]): the measure vanishes on meagre Baire sets, equivalently lives on the ultrafilters closed under countable intersection, which are exactly the realised points. This is the measure-theoretic face of the *Loomis–Sikorski* representation — a σ-complete Boolean algebra is a σ-tribe of sets modulo a σ-ideal — the countable-join bookkeeping that makes the equivalence go through. There is *no* OML analogue of any of this, and supplying one — or proving none exists — is the open problem of §5: the Boolean proof has an engine; the OML proof needs one and does not have it.

Remark 2.11 (Why the bridge breaks: meet-closed but not join-prime)*. The slogan: in the Boolean world "realised" bundles two properties that quantum non-distributivity pulls apart.* The Boolean bridge fuses two conditions on a realised point x: being *closed under countable meets* and being *off the countable-join defect* ($x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ for some n). Rao–Rao's "vanishes on meagre sets" is exactly the coincidence of the two. In a non-distributive OML they *split*: a principal filter $x = \uparrow p$ is always closed under countable meets, yet may fail join-primeness, since $p \leq \bigvee_n a_n$ does not force $p \leq a_n$ for any n (take $a_n = \mathrm{span}(e_n)$ in $L(H)$ and $p = \mathrm{span}(v)$ for *any* line not aligned with a basis vector — already $v = e_1 + e_2$: $p \leq \bigvee_n a_n = H$ but $p \not\leq a_n$ for every n). Such an x is a *realised* point that sits inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically $\uparrow p$ is an *isolated* point of $S_0(L(H))$, so D is non-meagre — the reverse of the Boolean case, where the defect is nowhere dense. This is the finitary wall of Remark 2.10 seen on the dual: the category route (clopens modulo the meagre ideal) chokes on the same D as the measure route, and in fact reduces to the MacNeille completion of §4.3(ii).

3 Two axes: extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two **independent** axes that the Boolean case fuses. A state s on A faces:

	Extension axis	**Descent axis**
Question	Does μ extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$?	Once extended, does the measure concentrate on the physical points $P(A)$?
Governed by	the meet-zero/orthogonal gap (§2.1)	σ-additivity — needs the missing engine (§2.4)
Status	**settled**: Horn–Tarski / Pitowsky feasibility; finite case = decidable LP; can fail for every state.	**open**. *The descent residue.*

Two observations pin down the extension axis.

Proposition 3.1 (Finite case)**.** *For finite A the extension condition is a decidable linear program (Horn–Tarski [20] / Pitowsky correlation-polytope feasibility [29]); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \le 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.8).*

Proposition 3.2 (Infinite $L(H)$, all states)**.** *Let* $\dim H = \infty$*.* No *state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof. (The idea: hide a copy of MO_2 *inside $L(H)$ whose four atoms are all infinite-dimensional, so even singular states must charge them.)* Write $H = H_0 \otimes \mathbb{C}^2$ with H_0 infinite-dimensional, and fix an orthonormal basis e, f of $\mathbb{C}^2$. Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^\perp = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e{+}f), \quad a_2^\perp = H_0 \otimes \mathbb{C}(e{-}f).$$

These four (infinite-dimensional) subspaces form a copy of MO_2 inside $L(H)$: $a_1 \perp a_1^\perp$ and $a_2 \perp a_2^\perp$, each pair joining to H, while all four *cross* pairs are meet-zero (distinct lines in $\mathbb{C}^2$ give $a_i \cap a_j = \mathbf{0}$) but *not* orthogonal. Orthoadditivity gives $s(a_i) + s(a_i^\perp) = s(H) = 1$ for $i = 1, 2$ — this uses only $s(\mathbf{1}) = 1$ and additivity on orthogonal pairs, so it holds for *every* state — whence $\sum s(a_i) = 2$ over the pairwise-meet-zero family $\{a_1, a_1^\perp, a_2, a_2^\perp\}$. Since h preserves meets ($h(a \wedge b) = h(a) \cap h(b)$, the filters being meet-closed), meet-zero elements map to disjoint clopens, so any charge forces $\sum s(a_i) \le 1$. Contradiction. The construction is finite ($n = 4$) and σ-completeness of the lattice is irrelevant. □

The MO_2 count is folklore (orthoadditivity plus $s(\mathbf{1}) = 1$; cf. [21]), and the $\sum s(a_i) = 2$ is already state-independent in $L(\mathbb{C}^2)$. What it buys *here* is different: realising MO_2 with *infinite-dimensional* atoms makes the obstruction reach the states that vanish on finite rank. In $L(\mathbb{C}^2)$ the four atoms are lines (finite rank), and a singular state — one with $s(e) = 0$ for every finite-dimensional e — assigns them all 0, so the count, though valid, says nothing there; it constrains only states positive on finite rank. Infinite-dimensional H_0 keeps each atom infinite-dimensional, so $s(a_i) + s(a_i^\perp) = 1$ holds regardless of behaviour on finite rank — dropping the finite-rank precondition of the line argument (Remark 3.3) and catching the singular states as well. This is the role of infinite-dimensionality, and it is what closes the extension axis on $L(H)$ completely.

Remark 3.3 (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the MO_2 family. If $s(e) > 0$ on some finite-dimensional e, it is positive on a 2-plane e_0; choosing k pairwise-non-orthogonal lines $p_1, \dots, p_k \subset e_0$ gives $2k$ distinct pairwise-meet-zero lines with $s(p_i) + s(p_i^\perp) = s(e_0)$, so a charge forces $k\,s(e_0) \le 1$ for all k — impossible once $s(e_0) > 0$. Lifting s to a functional via Mackey–Gleason / Bunce–Wright [2] ($B(H)$ has no type I_2 summand) and applying the Takesaki normal/singular decomposition identifies the states it reaches — those with nonzero normal part, i.e. $s(e) > 0$ on some finite-dimensional e, including every normal (Gleason) state — and the states it misses, the purely singular ones ($s(e) = 0$ on all finite e; e.g. ultrafilter vector states or Calkin-algebra pullbacks). Proposition 3.2 subsumes this case; the line argument is recorded because its technique is independent and because it makes precise where the singular states sit. The open problem is on the descent axis (§5).

4 What is known

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that question — the Boolean benchmark such an analogue would have to match, the partial OML results, and the routes already known to be blocked.

4.1 The Boolean benchmark

(The target an OML engine would have to hit.) In the Boolean case the Kelley–Vladimirov–Pták characterisation (Fremlin [12], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it is Dedekind σ-complete, weakly (σ, ∞)-distributive, and chargeable. Each condition has *no* clean OML analogue: weak (σ, ∞)-distributivity relies on distributivity of $\bigwedge$ over $\bigvee$; chargeability has no structural characterisation on OMLs.

4.2 Positive and obstructive OML results

The positive results require structure rich enough to approximate Hilbert-space geometry: Gleason [14] ($L(H)$, $\dim \ge 3$; σ-additivity *on the lattice*), Bunce–Wright [2] (JBW projection lattices), Chetcuti–Dvurečenskij [5]. The obstructive results show the Boolean machinery cannot be transplanted: Pták–Pulmannová [31] (a *finitary* measure condition — *subadditive*-unitality, $s(a \vee b) \le s(a) + s(b)$ with no countable joins — collapses the lattice to Boolean; their Theorem 1 is lattice-only, and the orthomodular-*poset* version fails, by Müller's set-representable counterexample [27]), Navara [28] (regularity does not force σ-additivity), Pták [30] (exotic state spaces via Greechie pasting).

4.3 Three blocked routes to a σ-OML engine

The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all blocked.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [9] and monotone σ-effect algebras *with* the Riesz Decomposition Property [10]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [21]), so the hypothesis that would buy the representation collapses the class:

$$\mathrm{OML} + \mathrm{RDP} \iff \text{Boolean}.$$

(ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [17]); one cannot complete to absorb countable joins. The topological version of this route — clopens of $S_0(A)$ modulo the meagre ideal — is the same completion in disguise (the regular-open algebra), and collapses non-distributivity (Remark 2.11).

(iii) **σ-Stone duality.** None exists. McDonald–Bimbó is finitary (Remark 2.10); Freytes' theory [13] of σ-complete OMLs is equational, with no set/tribe representation.

4.4 Prior art on point-free quantum probability

Every existing point-free or directed-context construction lands on one side of the σ-additivity-on-a-non-distributive-structure line.

Gleason tradition Varadarajan [35], Kalmbach [21], Pták–Pulmannová [31]: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009 [8] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS) [18, 19] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations Vickers [36], Simpson [33], Coquand–Spitters [6]: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities McDonald–Bimbó [26] and Cannon–Döring [4]: purely structural, no measure, no σ-version.

The unoccupied conjunction — point-free $\times$ σ-additive $\times$ natively non-distributive $\times$ directed-system-built — is exactly the descent residue. (The directed-context-yields-non-distributivity *idea* is itself not new; it is the shape of Bohrification and of colimit-over-Boolean-contexts generation of OMLs. Novelty would lie in the σ-additive-point-free-native combination.)

Remark 4.1 (The boundary of the negative results). Two qualifications sharpen what is and is not ruled out. *(i) No impossibility theorem* covers the *concrete* class: the known no-go results are finite or two-valued, and the one positive occupant $L(H)$+Gleason is *non-concrete* (Kochen–Specker [22], dim $\geq$ 3: $L(H)$ admits no two-valued states at all, a fortiori none order-determining), so concreteness is the qualifier on which the open frontier turns. *(ii) The absence of a σ-Loomis–Sikorski* is visible in the dualities themselves: both take infinite joins as a *closure*, not a union (Cannon–Döring $\mathrm{cls}(\bigcup S_i)$; McDonald–Bimbó $(\bigcup h(a_n))^{\perp\perp}$). One should not overstate this: Gudder's concrete logics [15] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

5 The open problem

Open Problem. Does a σ-complete, concrete, non-Boolean, infinite OML admit a Loomis–Sikorski-type representation (a σ-tribe of sets modulo a σ-ideal), or a countable-join-preserving σ-Stone duality — or can one prove that none exists?

Remark 5.1 (Whether the class is inhabited). A prior question is whether the class contains an example that exercises the descent axis at all. A member is descent-relevant only if it carries non-trivial countable orthogonal structure (infinite pairwise-orthogonal families, so that

σ-additivity is more than finite additivity) and meet-zero $\neq$ orthogonal pairs. The completeness the measure question requires is therefore σ-*ortho*completeness (countable *orthogonal* joins), not full σ-completeness, which is the stronger structural notion that the Loomis–Sikorski representation manipulates. The obvious candidates do not qualify: MO_κ (κ infinite) is concrete, infinite, non-Boolean and complete, but its maximal pairwise-orthogonal family of nonzero elements has only two members ($a \perp b$ among atoms iff $a = b^\perp$), so descent is vacuous, and it carries the MO_2 obstruction (no state extends, as on $L(H)$).

The closest candidate is Navara's σ-orthocomplete lattice logic [28]: infinite, non-Boolean, and σ-orthocomplete, with rich countable orthogonal structure, yet non-concrete, since its Greechie-stateless building blocks leave too few two-valued states to be order-determining. Substituting a concrete, stateful, non-Boolean block (MO_2) for Navara's stateless one yields concreteness for free, since sub-OMPs and products of concrete logics are concrete; and σ-orthocompleteness *survives* the substitution, since Navara's closure argument ([28, p. 428]) is state-free — the one step that uses the block, "$f_j(m) = 0$ and $f_i \perp f_j$ force f_j constant ($= 0$) on the block," carries verbatim for MO_2, where the only elements below $a^\perp$ are $\{0, a^\perp\}$. The swap is thus a genuine member of the class of the Problem (concrete, σ-orthocomplete, infinite, non-Boolean), so the class *is* inhabited as stated.

The sharper question is whether any such member is descent-relevant in the strong sense that the non-distributivity and the infinite orthogonal structure interleave, rather than segregating into a Boolean support skeleton carrying merely decorative non-Boolean blocks. The discriminating condition is *element-versus-family*:

$$\exists\, p \text{ and an } \textit{infinite} \text{ orthogonal family } \{a_n\} \text{ with } p \wedge a_n = 0 \ \forall n, \ \text{ yet } p \not\perp a_n. \tag{2}$$

(Within a single orthogonal family this never occurs — such a family generates a Boolean subalgebra, where meet-zero *is* orthogonality — so the interleaving must be of one element *against* the family.) On $L(H)$, (2) holds: for an orthonormal basis $\{e_n\}$ and a skew vector $f = \sum_n c_n e_n$ with all $c_n \neq 0$, the ray P_f meets each P_{e_n} at 0 yet is orthogonal to none. On MO_κ it fails vacuously (no infinite orthogonal family). For the MO_2-swap the condition is the *outstanding by-hand check*: an element supported in finitely many blocks is meet-zero-not-orthogonal to only finitely many of the cross-block a_n, so (2) requires the constancy condition to admit an element in MO_2-position in infinitely many blocks at once. If it cannot, the swap is segregated — a member of the as-stated class but not a strong-descent witness — and $L(H)$ remains the only structure known to satisfy (2), with no product/sublogic construction able to manufacture it (this sharpens, rather than replaces, the one-example status of the richness-obstructs-concreteness heuristic: $L(H)$ alone, by Kochen–Specker, both satisfies (2) and is non-concrete). Two qualifications carry over: the relevant completeness is σ-orthocompleteness, not σ-completeness; and $L(H)$ and Navara fail concreteness for different reasons (Kochen–Specker colouring versus an imported stateless block). Details of the substitution and the (2) reduction are recorded in the inhabitation note.

A positive answer would furnish a relational probability theory without realisations: σ-additive probability built from the entailment relation of a non-distributive OML rather than descended from a sample space — the non-Boolean analogue of localic (pointless) measure theory. A negative answer, in the form of an impossibility theorem, would close the entire cluster, both residues, at once. The present state of the art constrains the problem from both sides: no impossibility theorem is known for the concrete class (Remark 4.1), while all three constructive routes to a σ-OML engine are blocked (§4.3). What is required is therefore not a new idea in the abstract but a representation that bypasses all three routes.

Remark 5.2 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). The motivating idea ("probability from relations, point-free") may *look* delivered by $L(H)$ + Gleason. It is not, and the reason is a equivocation between two point-spaces: (A) the Hilbert rays of H, from which the Gleason density ρ in $s = \mathrm{tr}(\rho\,\cdot)$ is built, and (B) the McDonald–Bimbó dual filters $P(A)$. The programme defines *realisation* as concentration on (B). Gleason removes (B) but *requires* (A) — and is a *representation* theorem, reducing every lattice state to the point datum ρ, the most point-ful result available. So $L(H)$/Gleason exhibits a *separation* — σ-additive measures exist on the lattice (Gleason, for the normal states), while (by Proposition 3.2) no state of any kind even extends to a charge on $S_0(L(H))$, so a fortiori none concentrates on the dual points $P(A)$ — not point-freeness, and it cannot serve as the existence proof sought. Because it cannot, the σ-OML representation above is the *only* candidate engine; it would be an error to regard Gleason's theorem as already furnishing a relational, point-free probability.

6 Directions

With the extension axis closed on $L(H)$ (Proposition 3.2), a single direction remains.

The general construction

Construct a Loomis–Sikorski-type representation, or a countable-join-preserving σ-Stone duality, for a concrete non-Boolean infinite σ-OML — or prove that none exists. This resolves the open problem directly, but no partial construction is currently in hand. This is the *descent*-axis residue: the obstruction is the failure of h to preserve countable joins (Remark 2.10), not the orthogonal/meet-zero gap, and so Proposition 3.2 does not reach it.

The other candidate direction — the singular case for $L(H)$ — is now closed, and the way it closed is instructive. One might hope a *singular* state would extend even though no normal state does, furnishing the first explicit "extends but may fail to concentrate" example the general construction lacks. It does not: Proposition 3.2 catches every state, singular included. The reduction that settled it is worth recording, since the natural expectation was that the singular case would require infinitary methods.

Remark 6.1 (Finite versus countable: finite suffices). One could ask whether the obstruction to extending a singular state is realised by a *finite* or only by a *countable* pairwise-meet-zero family. The line argument of Remark 3.3 is finitary but vacuous on singular states ($s(e) = 0$ on every finite-dimensional e), which might suggest the singular obstruction needs a countable family and is therefore entangled with the descent-axis passage. It does not: the four infinite-dimensional subspaces of Proposition 3.2 are a *finite* ($n = 4$) pairwise-meet-zero family forcing $\sum_i s(a_i) = 2$, and the forcing is state-independent. So the singular case is a self-contained, finitary problem, separable from the general construction — not governed by the finitary-to-σ passage at all. The line argument is vacuous on singular states; the MO_2-with-infinite-dimensional-atoms argument is not, and that is the whole difference.

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